



COMP9517 Computer Vision
Example Exam Questions

Instructions

- Time allowed: 10 minutes reading + 3 hours working.
- The exam will count for 50% of your final course mark.
- The maximum achievable marks for each question are as indicated.
- All questions may be attempted (non-attempted questions do not attract marks).
- Numerical answers should be given correct to two decimal places unless otherwise noted.
- For multiple choice questions you must select the one best answer.
- No materials may be taken into or out of the examination room.
- It is not allowed (and not possible) to consult any works or the internet.
- Do not log out until you have saved and submitted your answers.

The exam consists of two parts:
Part A with multiple-choice questions for a total of 20 points.
Part B with open questions for a total of 30 points.
Together, the total number of achievable points is 50.
Below are 10 example questions for each part.

Part A: Multiple-Choice Questions

Question (1 mark): What is the purpose of convolutional layers in CNNs?

- A) Reducing the size of the feature maps.
B) Calculating the dot product of the input and kernels.
C) Applying a nonlinear activation function.
D) Making the network learn faster.

Question (1 mark): What does transfer learning with CNNs involve?

- A) Training a model from scratch for each new task.
B) Using a pretrained model and fine-tuning it for a new task.
C) Converting image data to text data to facilitate learning.
D) Combining multiple CNNs into a single model for better performance.

Question (1 mark): In many image processing operations, the image must be expanded a little bit to solve the boundary problem. What type of expansion is implicitly used by the Fourier transform?

- A) Padding
B) Clamping
C) Wrapping
D) Mirroring

Question (1 mark): Which one of the following equations concerning convolution ($\ast$) of two images (f and g) is correct?

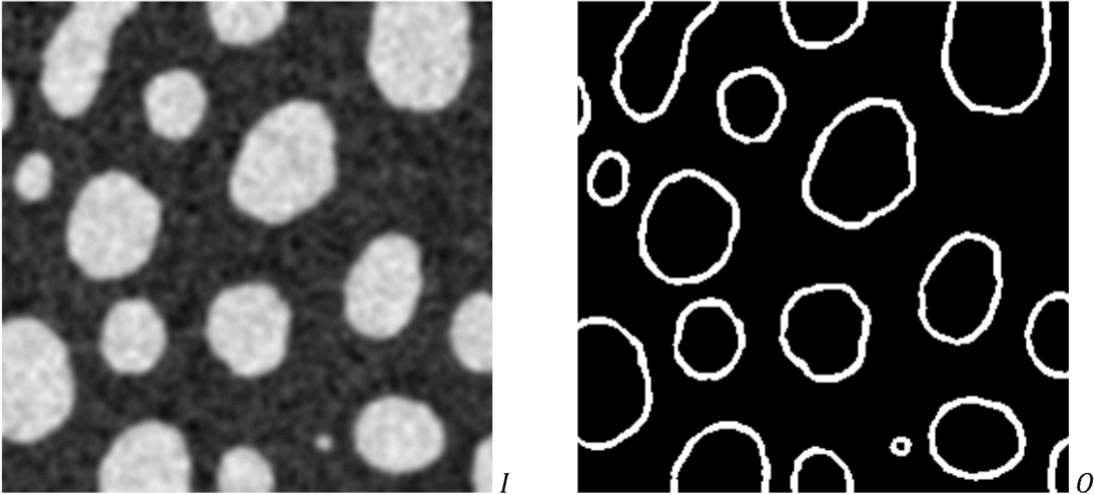
- A) $f \ast g = (g \ast f)^{-1}$
B) $f \ast (g + h) = (f \ast g) + (f \ast h)$
C) $a \cdot (f \ast g) = (a \cdot f) \ast (a \cdot g)$
D) $f \ast g = f \ast g$

Question (1 mark): What is the effect of the following 2D convolution kernel on an image?

0	1	0
1	-4	1
0	1	0

- A) It approximates the sum of first-order derivatives in x and y .
B) It approximates the sum of second-order derivatives in x and y .
C) It approximates the product of first-order derivatives in x and y .
D) It approximates the product of second-order derivatives in x and y .

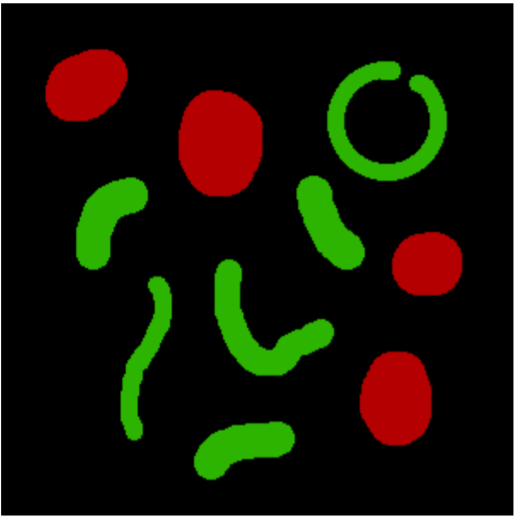
Question (1 mark): Suppose we have the following gray-scale input image I (shown on the left) and we would like to have a binary output image O (as shown on the right) containing the edges of the objects in the input. Also suppose that the only image processing operations available for this are intensity thresholding, dilation and erosion (binary and gray-scale), and image subtraction. Which algorithm will not produce the desired output?



- A) Thresholding (result = T), then erosion (result = E), then subtraction (result = $E - T$).
B) Erosion (result = E), then subtraction (result = $I - E$), then thresholding (result = T).
C) Thresholding (result = T), then dilation (result = D), then subtraction (result = $D - T$).

D) Dilation (result = D), then subtraction (result = $D - I$), then thresholding (result = T).

Question (1 mark): Given the image below showing the result of a segmentation of various objects and the desired classification of these objects. The two different colours (red and green) indicate the two different classes into which the objects are to be divided. A straightforward way to do classification is by computing the value of a quantitative shape measure for each object and then thresholding those values. Suppose we compute the circularity and the eccentricity. Which of these two measures can be used to produce the shown classification?



- A) Only circularity.
B) Only eccentricity.
C) Both circularity and eccentricity.
D) Neither circularity nor eccentricity.

Question (1 mark): Given the image below containing about two dozen algae (the bright spots) which we would like to segment using an automatic thresholding method. Which automatic thresholding method will give the best segmentation result here?



- A) Mean thresholding.
B) IsoData thresholding.
C) Otsu thresholding.
D) Triangle thresholding.

Question (1 mark): Which one of the following statements on image filtering is incorrect?

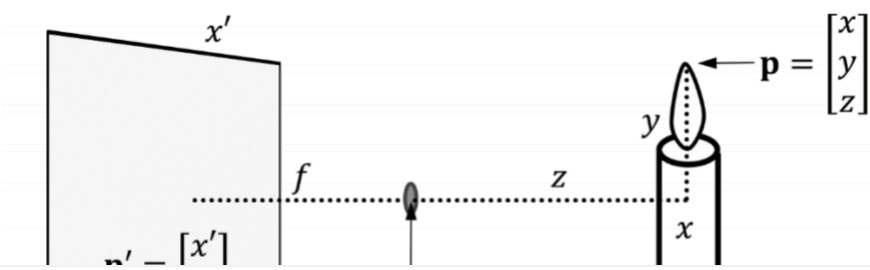
- A) Median filtering reduces noise in images.
B) Low-pass filtering smooths fine details in images.
C) High-pass filtering results in blurry images.
D) Notch filtering removes specific image frequencies.

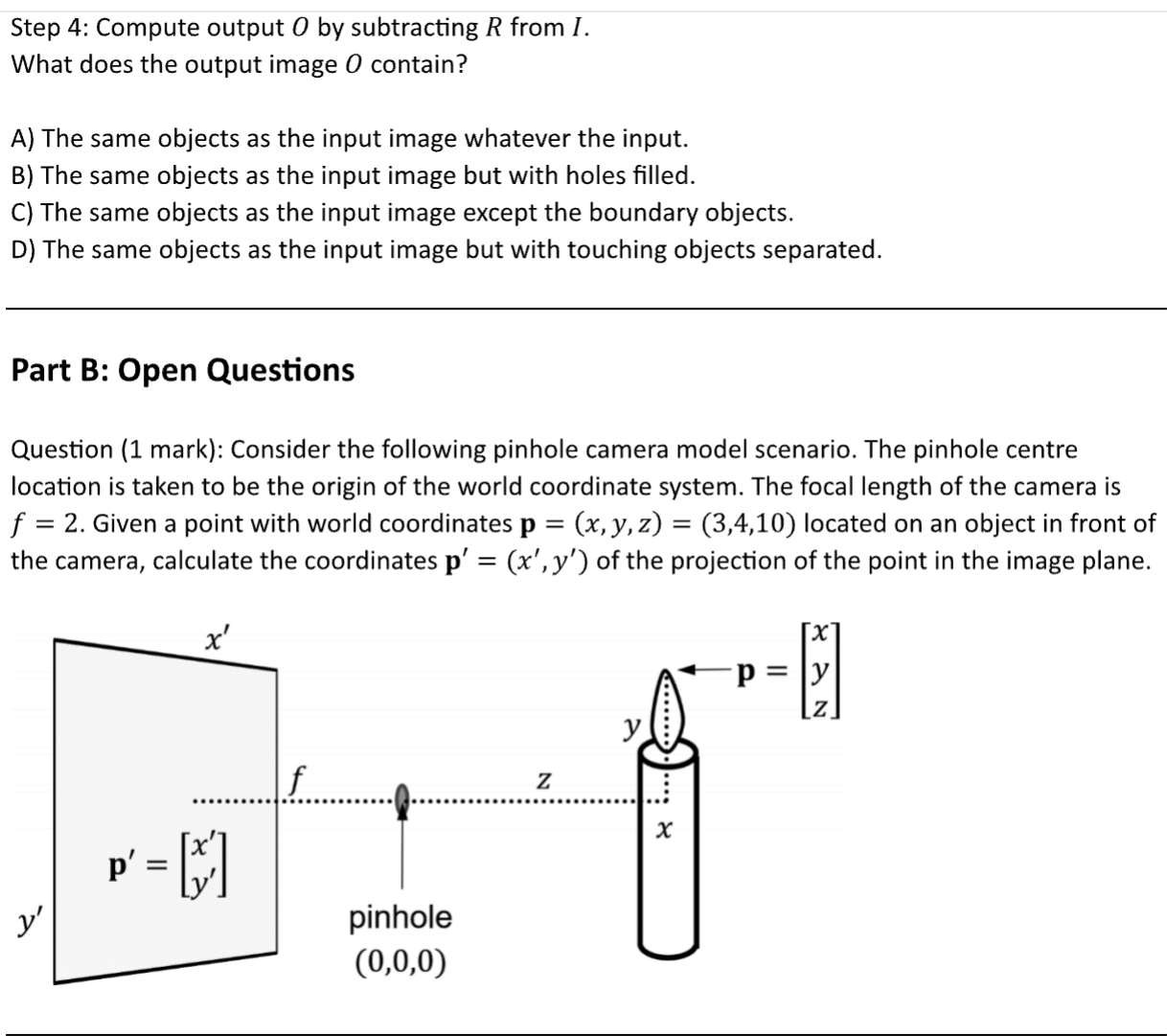
Question (1 mark): Given a binary input image I to which we apply the following algorithm:
Step 1: Create a copy C of input image I .
Step 2: Copy the boundary pixels of C into new image B .
Step 3: Compute the reconstruction R of C from B .
Step 4: Compute output O by subtracting R from I .
What does the output image O contain?

- A) The same objects as the input image whatever the input.
B) The same objects as the input image but with holes filled.
C) The same objects as the input image except the boundary objects.
D) The same objects as the input image but with touching objects separated.

Part B: Open Questions

Question (1 mark): Consider the following pinhole camera model scenario. The pinhole centre location is taken to be the origin of the world coordinate system. The focal length of the camera is $f = 2$. Given a point with world coordinates $\mathbf{p} = (x, y, z) = (3, 4, 10)$ located on an object in front of the camera, calculate the coordinates $\mathbf{p}' = (x', y')$ of the projection of the point in the image plane.





The diagram illustrates the hierarchical structure of the proposed CNN for pedestrian detection. It shows a **Cell (5x8 pixels)** which is a 5x8 grid of pixels. A **Block (2x2 Cells)** is a 2x2 grid of cells. A **Detection Window (7x15 Blocks)** is a 7x15 grid of blocks, containing a stick figure of a pedestrian. A **Cell Histogram** is shown below the cell, representing the distribution of cell features.

Question (2 marks): An important hyperparameter in performing convolutions in a CNN is the stride.

(i) What is the difference between a stride of 1 and a stride of 2 in a convolutional layer? (1 mark)

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(ii) How do these stride values impact the size of the output feature maps? (1 mark)

Question (2 marks): Spatial image filtering by convolution is a linear, shift-invariant operation.

(i) Explain the meaning of linearity of convolution. (1 mark)

(ii) Explain the meaning of shift-invariance of convolution. (1 mark)

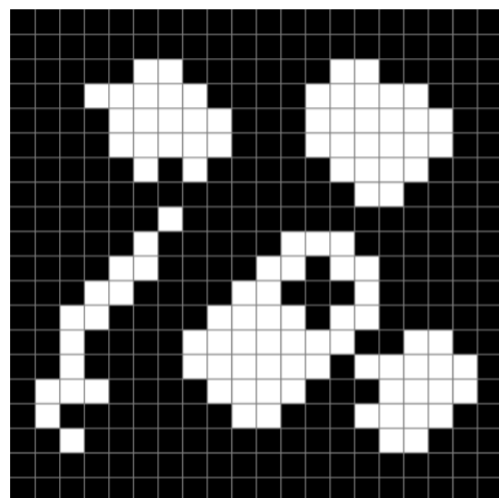
Question (2 marks): Suppose we have a 2D image that is rather noisy and we want to smooth it by filtering with a Gaussian kernel of size 3×3 voxels. How many floating-point operations (additions and multiplications combined) does it take per pixel to perform the filtering under the following conditions? Give the number and show how you computed it.

(i) The convolution with the kernel is implemented in a nonseparable way. (1 mark)

(ii) The convolution with the kernel is implemented in a separable way. (1 mark)

Question (2 marks): Suppose we are given a 2D image f and we want to compute the edge image $\|\nabla f\|$. Describe which image processing operations are required to compute the edge image.

Question (1 mark): Given the following binary image after a segmentation. To automatically identify the objects (in white) in this image we can use the connected components labelling algorithm. How many separate objects will this algorithm find here if it uses 4-connectivity?



Question (1 mark): Consider two 8-bit images whose gray levels span the full range from 0 to 255. If we repeatedly add the second image to the first, what is the limiting effect?

Question (2 marks): Given the following 2D convolution kernel.

1	0	-1
2	0	-2
1	0	-1

(i) What is the name of this kernel? (1 mark)

(ii) What effect does this kernel have on an image?